Take-Home Exercise for Research Scientist Position

Please complete the following practical within 24-hrs, at the end of the 24-hours window you will have to submit your work and present it in a 30 min research-deep-dive. Make sure your solutions are clear, concise, and well-documented. This will help us understand your thought process and technical abilities. Good luck!

Objective: Build a model predicting likelihood for lease renewal.

Instructions:

1. **Data:**
   - You will have access to two main forms of data:
     i. JSON files containing 1000 conversations of tenants considering renewing their leases;
     ii. A SQLite database consisting of three types of tables accessible at this link. The tables are summarized as follows:
        1. **residents**: resident information
        2. **maintenance_history**: maintenance work order submissions by resident
        3. **renewal_offers**: renewal offers provided to each unit
   - Both i and ii are split between training and test.

2. **Task 1 - Build a Renewal Prediction Model**
   - Build a model that uses the available data to predict whether or not a unit/resident renews
   - Discuss any preprocessing steps
3. **Task 2 - Model Evaluation:**
   - Identify a metric that demonstrates your model's performance compared to a random baseline.
   - Output a csv with predictions for each resident in the test dataset. The csv should have three mandatory columns: `resident_id`, a `unit_number` and `renewal_decision`, please add `probability_of_renewal` if your model allows. We will run our internal eval on the test set.
4. **Task 3 - Reason Categorization:**
   - Using the `gte-base` sentence transformer model, train a model to categorize the reasons tenants do not renew their leases.
   - Analyze the reasons given by tenants and come up with appropriate categories that encapsulate the range of reasons for not renewing.

    - Discuss how you defined these categories and any patterns or insights observed from the available data.
5. **Presentation of Work:**
    - Prepare a short presentation highlighting the high-level approach and results for the research deep-dive.
    - Have a Jupyter notebook ready - which will be part of the submission - and which will be used during the research deep-dive. Make sure the notebook contains clean and well documented code.